

Challenges related to Ducts, Chimneys and Tail-end Alignment

Paint deposition measurement & cleaning inside the ducts

Deposition of paint inside the exhaust duct of painting stations is a critical safety issue as it may lead to a significant fire hazard. Paint exhaust ducts are of MS material sheets which are rectangular with cross sections ($\sim 2 \times 1.5$ meters) and paint gets deposited on the walls, over a period. Currently there is no method to measure the quantity of paint deposited on exhaust ducts while the painting process is on. It is important to know the quantity of paint deposited so cleaning can be initiated thus reducing the risk of a fire hazard.

We are seeking a measurement of paint deposition on the duct walls without opening up or sending people inside the ducts to measure and also clean the paint layers that are deposited on the ducts when it exceeds the defined levels.